

Pure Mathematics 2 Practice Paper J8 **MARK SCHEME**

Question 1

Question Number	Scheme	Marks
	(a) $x = 1 - 2y^3 \Rightarrow y = \left(\frac{1-x}{2}\right)^{\frac{1}{3}} \text{ or } \sqrt[3]{\frac{1-x}{2}}$ $f^{-1}: x \mapsto \left(\frac{1-x}{2}\right)^{\frac{1}{3}}$ Ignore domain	M1 A1 (2)
	(b) $gf(x) = \frac{3}{1-2x^3} - 4$ $= \frac{3-4(1-2x^3)}{1-2x^3}$ $= \frac{8x^3-1}{1-2x^3} *$ $gf: x \mapsto \frac{8x^3-1}{1-2x^3}$ Ignore domain	M1 A1 M1 cso A1 (4)
	(c) $8x^3 - 1 = 0$ $x = \frac{1}{2}$ Attempting solution of numerator = 0 Correct answer and no additional answers	M1 A1 (2)

Question 2

Question Number	Scheme	Marks
(a)	** represents a constant (which must be consistent for first accuracy mark) $(8-3x)^{\frac{1}{3}} = \underline{(8)^{\frac{1}{3}} \left(1 - \frac{3x}{8}\right)^{\frac{1}{3}}} = \underline{2 \left(1 - \frac{3x}{8}\right)^{\frac{1}{3}}}$ Takes 8 outside the bracket to give any of $\underline{(8)^{\frac{1}{3}}}$ or $\underline{2}$. Expands $(1+**x)^{\frac{1}{3}}$ to give a simplified or an un-simplified $1 + \left(\frac{1}{3}\right)(**x)$; A correct simplified or an un-simplified {.....} expansion with candidate's followed through $(**x)$ with $** \neq 1$ $= 2 \left\{ 1 + \left(\frac{1}{3}\right)(**x) + \frac{\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)}{2!} (**x)^2 + \frac{\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)\left(-\frac{5}{3}\right)}{3!} (**x)^3 + \dots \right\}$ Award SC M1 if you see $\frac{\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)}{2!} (**x)^2 + \frac{\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)\left(-\frac{5}{3}\right)}{3!} (**x)^3$ $= 2 \left\{ 1 + \left(\frac{1}{3}\right)\left(-\frac{3x}{8}\right) + \frac{\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)}{2!} \left(-\frac{3x}{8}\right)^2 + \frac{\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)\left(-\frac{5}{3}\right)}{3!} \left(-\frac{3x}{8}\right)^3 + \dots \right\}$ Either $2\left\{1 - \frac{1}{8}x \dots\right\}$ or anything that cancels to $2 - \frac{1}{4}x$; Simplified $-\frac{1}{32}x^2 - \frac{5}{768}x^3$ $= 2 \left\{ 1 - \frac{1}{8}x; -\frac{1}{64}x^2 - \frac{5}{1536}x^3 - \dots \right\}$ $= 2 - \frac{1}{4}x; -\frac{1}{32}x^2 - \frac{5}{768}x^3 - \dots$	B1 M1; A1 $\checkmark$ A1; A1 [5]
(b)	$(7.7)^{\frac{1}{3}} \approx 2 - \frac{1}{4}(0.1) - \frac{1}{32}(0.1)^2 - \frac{5}{768}(0.1)^3 - \dots$ $= 2 - 0.025 - 0.0003125 - 0.0000065104166\dots$ $= 1.97468099\dots$ Attempt to substitute $x = 0.1$ into a candidate's binomial expansion. awrt 1.9746810	M1 A1 [2]
		7 marks

You would award B1M1A0 for

$$= 2 \left\{ 1 + \left(\frac{1}{3}\right)\left(-\frac{3x}{8}\right) + \frac{\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)}{2!} \left(-\frac{3x}{8}\right)^2 + \frac{\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)\left(-\frac{5}{3}\right)}{3!} (-3x)^3 + \dots \right\}$$

because ** is not consistent.

If you see the constant term "2" in a candidate's final binomial expansion, then you can award B1.

Be wary of calculator value of $(7.7)^{\frac{1}{3}} = 1.974680822\dots$

Question Number	Scheme	Marks
Aliter (a) Way 2	$(8-3x)^{\frac{1}{3}}$ $= \left\{ \begin{aligned} &(8)^{\frac{1}{3}} + \left(\frac{1}{3}\right)(8)^{-\frac{2}{3}}(**x) + \frac{\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)}{2!}(8)^{-\frac{5}{3}}(**x)^2 \\ &+ \frac{\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)\left(-\frac{5}{3}\right)}{3!}(8)^{-\frac{8}{3}}(**x)^3 + \dots \end{aligned} \right\}$ with $** \neq 1$ $= \left\{ \begin{aligned} &(8)^{\frac{1}{3}} + \left(\frac{1}{3}\right)(8)^{-\frac{2}{3}}(-3x) + \frac{\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)}{2!}(8)^{-\frac{5}{3}}(-3x)^2 \\ &+ \frac{\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)\left(-\frac{5}{3}\right)}{3!}(8)^{-\frac{8}{3}}(-3x)^3 + \dots \end{aligned} \right\}$ $= \left\{ 2 + \left(\frac{1}{3}\right)\left(\frac{1}{4}\right)(-3x) + \left(-\frac{1}{9}\right)\left(\frac{1}{32}\right)(9x^2) + \left(\frac{5}{81}\right)\left(\frac{1}{256}\right)(-27x^3) + \dots \right\}$ $= 2 - \frac{1}{4}x - \frac{1}{32}x^2 - \frac{5}{768}x^3 - \dots$	2 or $(8)^{\frac{1}{3}}$ (See note ↓) B1 Expands $(8-3x)^{\frac{1}{3}}$ to give an un-simplified or simplified M1; $(8)^{\frac{1}{3}} + \left(\frac{1}{3}\right)(8)^{-\frac{2}{3}}(**x)$; A correct un-simplified or simplified {.....} expansion with A1 ✓ candidate's followed through $(**x)$ Award SC M1 if you see $\frac{\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)}{2!}(8)^{-\frac{5}{3}}(**x)^2$ $+ \frac{\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)\left(-\frac{5}{3}\right)}{3!}(8)^{-\frac{8}{3}}(**x)^3$ Anything that cancels to $2 - \frac{1}{4}x$; A1; or $2\left\{1 - \frac{1}{8}x \dots\right\}$ Simplified $-\frac{1}{32}x^2 - \frac{5}{768}x^3$ A1 [5]

Be wary of calculator value of $(7.7)^{\frac{1}{3}} = 1.974680822\dots$

If you see the constant term "2" in a candidate's final binomial expansion, then you can award B1.

Takes out brackets.

Writing down $\frac{1}{(t+1)(t+2)} = \frac{1}{(t+1)} + \frac{1}{(t+2)}$ means first M1A0 in (b).

Writing down $\frac{1}{(t+1)(t+2)} = \frac{1}{(t+1)} - \frac{1}{(t+2)}$ means first M1A1 in (b).



Question 3

Question Number	Scheme	Marks
	(a) $\frac{dy}{dx} = 2e^{2x} \tan x + e^{2x} \sec^2 x$ $\frac{dy}{dx} = 0 \Rightarrow 2e^{2x} \tan x + e^{2x} \sec^2 x = 0$ $2 \tan x + 1 + \tan^2 x = 0$ $(\tan x + 1)^2 = 0$ $\tan x = -1 \quad *$ cso	M1 A1+A1 M1 A1 A1 (6)
	(b) $\left(\frac{dy}{dx} \right)_0 = 1$ Equation of tangent at $(0, 0)$ is $y = x$	M1 A1 (2) [8]

Question 4

Question Number	Scheme	Marks
(a)	$x^3 - 4y^2 = 12xy \quad (\text{eqn } *)$ $x = -8 \Rightarrow -512 - 4y^2 = 12(-8)y$ $-512 - 4y^2 = -96y$ $4y^2 - 96y + 512 = 0$ $y^2 - 24y + 128 = 0$ $(y - 16)(y - 8) = 0$ $y = \frac{24 \pm \sqrt{576 - 4(128)}}{2}$ $y = 16 \text{ or } y = 8.$	Substitutes $x = -8$ (at least once) into * to obtain a three term quadratic in y. Condone the loss of $= 0$. M1 An attempt to solve the quadratic in y by either factorising or by the formula or by <i>completing the square</i>. dM1 Both $y = 16$ and $y = 8$. or $(-8, 8)$ and $(-8, 16)$. A1 [3]
(b)	$\left\{ \frac{\cancel{dy}}{\cancel{dx}} \times \right\} 3x^2 - 8y \frac{dy}{dx} = \left(12y + 12x \frac{dy}{dx} \right)$ $\left\{ \frac{dy}{dx} = \frac{3x^2 - 12y}{12x + 8y} \right\}$ $@ (-8, 8), \quad \frac{dy}{dx} = \frac{3(64) - 12(8)}{12(-8) + 8(8)} = \frac{96}{-32} = -3,$ $@ (-8, 16), \quad \frac{dy}{dx} = \frac{3(64) - 12(16)}{12(-8) + 8(16)} = \frac{0}{32} = 0.$	Differentiates implicitly to include either $\pm ky \frac{dy}{dx}$ or $12x \frac{dy}{dx}$. Ignore $\frac{dy}{dx} = \dots$ M1 Correct LHS equation; Correct application of product rule A1; (B1) <i>not necessarily required.</i> Substitutes $x = -8$ and <i>at least one</i> of their y-values to attempt to find any one of $\frac{dy}{dx}$. dM1 One gradient found. A1 Both gradients of -3 and 0 <i>correctly</i> found. A1 cso [6]
		9 marks

Question Number	Scheme	Marks
Aliter (b) Way 2	$\left\{ \begin{array}{l} \cancel{3x^2} \times \\ \cancel{12y} \end{array} \right\} 3x^2 \frac{dx}{dy} - 8y; = \left(\underline{12y \frac{dx}{dy} + 12x} \right)$ $\left\{ \frac{dy}{dx} = \frac{3x^2 - 12y}{12x + 8y} \right\}$ $@ (-8, 8), \quad \frac{dy}{dx} = \frac{3(64) - 12(8)}{12(-8) + 8(8)} = \frac{96}{-32} = \underline{-3},$ $@ (-8, 16), \quad \frac{dy}{dx} = \frac{3(64) - 12(16)}{12(-8) + 8(16)} = \frac{0}{32} = \underline{0}.$	Differentiates implicitly to include either $\pm kx^2 \frac{dx}{dy}$ or $12y \frac{dx}{dy}$. Ignore $\frac{dx}{dy} = \dots$ Correct LHS equation <u>Correct application of product rule</u> <i>not necessarily required.</i> Substitutes $x = -8$ and <i>at least one</i> of their y-values to attempt to find any one of $\frac{dy}{dx}$ or $\frac{dx}{dy}$. One gradient found. Both gradients of <u>-3</u> and <u>0</u> <i>correctly</i> found. M1 A1; (B1) dM1 A1 A1 cso [6]

Question Number	Scheme	Marks
Aliter (b) Way 3	$x^3 - 4y^2 = 12xy \text{ (eqn *)}$ $4y^2 + 12xy - x^3 = 0$ $y = \frac{-12x \pm \sqrt{144x^2 - 4(4)(-x^3)}}{8}$ $y = \frac{-12x \pm \sqrt{144x^2 + 16x^3}}{8}$ $y = \frac{-12x \pm 4\sqrt{9x^2 + x^3}}{8}$ $y = -\frac{3}{2}x \pm \frac{1}{2}(9x^2 + x^3)^{\frac{1}{2}}$ $\frac{dy}{dx} = -\frac{3}{2} \pm \frac{1}{2}\left(\frac{1}{2}\right)(9x^2 + x^3)^{-\frac{1}{2}}; (18x + 3x^2)$ $\frac{dy}{dx} = -\frac{3}{2} \pm \frac{18x + 3x^2}{4(9x^2 + x^3)^{\frac{1}{2}}}$ $@ x = -8 \quad \frac{dy}{dx} = -\frac{3}{2} \pm \frac{18(-8) + 3(64)}{4(9(64) + (-512))^{\frac{1}{2}}}$ $= -\frac{3}{2} \pm \frac{48}{4\sqrt{(64)}} = -\frac{3}{2} \pm \frac{48}{32}$ $\therefore \frac{dy}{dx} = -\frac{3}{2} \pm \frac{3}{2} = \underline{-3}, \underline{0}.$	A credible attempt to make y the subject and an attempt to differentiate either $-\frac{3}{2}x$ or $\frac{1}{2}(9x^2 + x^3)^{\frac{1}{2}}$. M1 $\frac{dy}{dx} = -\frac{3}{2} \pm k(9x^2 + x^3)^{-\frac{1}{2}}(g(x))$ A1 $\frac{dy}{dx} = -\frac{3}{2} \pm \frac{1}{2}\left(\frac{1}{2}\right)(9x^2 + x^3)^{-\frac{1}{2}}; (18x + 3x^2)$ A1 Substitutes $x = -8$ find any one of $\frac{dy}{dx}$. dM1 One gradient correctly found. A1 Both gradients of <u>-3</u> and <u>0</u> correctly found. A1

[6]



Question 5

Question Number	Scheme	Marks
	(a) $\frac{dy}{dx} = 6 \cos 2x - 8 \sin 2x$ $\left(\frac{dy}{dx}\right)_0 = 6$ $y - 4 = -\frac{1}{6}x$ or equivalent (b) $R = \sqrt{3^2 + 4^2} = 5$ $\tan \alpha = \frac{4}{3}, \alpha \approx 0.927$ awrt 0.927 (c) $\sin(2x + \text{their } \alpha) = 0$ $x = -2.03, -0.46, 1.11, 2.68$ First A1 any correct solution; second A1 a second correct solution; third A1 all four correct and to the specified accuracy or better. Ignore the y-coordinate.	M1 A1 B1 M1 A1 (5) M1 A1 M1 A1 (4) M1 A1 A1 A1 (4) [13]



Question 6

Question Number	Scheme	Marks
	(a) $f(2) = 0.38 \dots$ $f(3) = -0.39 \dots$ Change of sign (and continuity) $\Rightarrow$ root in $(2, 3)$ *	M1 A1 (2) cso
	(b) $x_1 = \ln 4.5 + 1 \approx 2.50408$ $x_2 \approx 2.50498$ $x_3 \approx 2.50518$	M1 A1 A1 (3)
	(c) Selecting $[2.5045, 2.5055]$, or appropriate tighter range, and evaluating at both ends. $f(2.5045) \approx 6 \times 10^{-4}$ $f(2.5055) \approx -2 \times 10^{-4}$ Change of sign (and continuity) $\Rightarrow$ root $\in (2.5045, 2.5055)$ $\Rightarrow$ root = 2.505 to 3 dp *	M1 A1 (2) cso [7]
	Note: The root, correct to 5 dp, is 2.50524	



Question 7

Question Number	Scheme	Marks
(i)	$\int \ln\left(\frac{x}{2}\right) dx = \int 1 \cdot \ln\left(\frac{x}{2}\right) dx \Rightarrow \left\{ \begin{array}{l} u = \ln\left(\frac{x}{2}\right) \Rightarrow \frac{du}{dx} = \frac{1}{x} = \frac{1}{x} \\ \frac{dv}{dx} = 1 \Rightarrow v = x \end{array} \right\}$ $\int \ln\left(\frac{x}{2}\right) dx = x \ln\left(\frac{x}{2}\right) - \int x \cdot \frac{1}{x} dx$ $= x \ln\left(\frac{x}{2}\right) - \int 1 dx$ $= x \ln\left(\frac{x}{2}\right) - x + c$	Use of 'integration by parts' formula in the correct direction. M1 Correct expression. A1 An attempt to multiply x by a candidate's $\frac{a}{x}$ or $\frac{1}{bx}$ or $\frac{1}{x}$. dM1 Correct integration with $+c$ A1 aef [4]
(ii)	$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin^2 x dx$ [NB: $\cos 2x = \pm 1 \pm 2\sin^2 x$ gives $\sin^2 x = \frac{1 - \cos 2x}{2}$] $= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{1 - \cos 2x}{2} dx = \frac{1}{2} \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (1 - \cos 2x) dx$ $= \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$ $= \frac{1}{2} \left[\left(\frac{\pi}{2} - \frac{\sin(\pi)}{2} \right) - \left(\frac{\pi}{4} - \frac{\sin(\frac{\pi}{2})}{2} \right) \right]$ $= \frac{1}{2} \left[\left(\frac{\pi}{2} - 0 \right) - \left(\frac{\pi}{4} - \frac{1}{2} \right) \right]$ $= \frac{1}{2} \left(\frac{\pi}{4} + \frac{1}{2} \right) = \frac{\pi}{8} + \frac{1}{4}$	Consideration of double angle formula for $\sin^2 x$ M1 Integrating to give $\pm ax \pm b \sin 2x$; Correct result of anything equivalent to $\frac{1}{2}x - \frac{1}{4}\sin 2x$ dM1 A1 Substitutes limits of $\frac{\pi}{2}$ and $\frac{\pi}{4}$ and subtracts the correct way round. ddM1 $\frac{1}{2} \left(\frac{\pi}{4} + \frac{1}{2} \right)$ or $\frac{\pi}{8} + \frac{1}{4}$ A1 aef Candidate must collect their π term and constant term together for A1 [5]
		9 marks

Question Number	Scheme	Marks
<i>Aliter</i> (i) Way 2	$\int \ln\left(\frac{x}{2}\right) dx = \int (\ln x - \ln 2) dx = \int \ln x dx - \int \ln 2 dx$ $\int \ln x dx = \int 1 \cdot \ln x dx \Rightarrow \left\{ \begin{array}{l} u = \ln x \Rightarrow \frac{du}{dx} = \frac{1}{x} \\ \frac{dv}{dx} = 1 \Rightarrow v = x \end{array} \right\}$ $\int \ln x dx = x \ln x - \int x \cdot \frac{1}{x} dx$ $= x \ln x - x + c$ $\int \ln 2 dx = x \ln 2 + c$ Hence, $\int \ln\left(\frac{x}{2}\right) dx = x \ln x - x - x \ln 2 + c$	Use of 'integration by parts' formula in the correct direction. M1 Correct integration of $\ln x$ with or without $+ c$ A1 Correct integration of $\ln 2$ with or without $+ c$ M1 Correct integration with $+ c$ A1 aef [4]

Question 8

Question Number	Scheme	Marks
	(a) $\cos(2x+x) = \cos 2x \cos x - \sin 2x \sin x$ $= (2\cos^2 x - 1)\cos x - (2\sin x \cos x)\sin x$ $= (2\cos^2 x - 1)\cos x - 2(1 - \cos^2 x)\cos x$ any correct expression $= 4\cos^3 x - 3\cos x$	M1 M1 A1 A1 (4)
	(b)(i) $\frac{\cos x}{1+\sin x} + \frac{1+\sin x}{\cos x} = \frac{\cos^2 x + (1+\sin x)^2}{(1+\sin x)\cos x}$ $= \frac{\cos^2 x + 1 + 2\sin x + \sin^2 x}{(1+\sin x)\cos x}$ $= \frac{2(1+\sin x)}{(1+\sin x)\cos x}$ $= \frac{2}{\cos x} = 2\sec x$ *	M1 A1 M1 cso A1 (4)
	(c) $\sec x = 2$ or $\cos x = \frac{1}{2}$ $x = \frac{\pi}{3}, \frac{5\pi}{3}$ accept awrt 1.05, 5.24	M1 A1, A1 (3)
[11]		

Question 9

Question Number	Scheme	Marks
(a)	$\left[x = \ln(t+2), y = \frac{1}{t+1} \right], \Rightarrow \frac{dx}{dt} = \frac{1}{t+2}$ $\text{Area}(R) = \int_{\ln 2}^{\ln 4} \frac{1}{t+1} dx; = \int_0^2 \left(\frac{1}{t+1} \right) \left(\frac{1}{t+2} \right) dt$ Changing limits, when: $x = \ln 2 \Rightarrow \ln 2 = \ln(t+2) \Rightarrow 2 = t+2 \Rightarrow t = 0$ $x = \ln 4 \Rightarrow \ln 4 = \ln(t+2) \Rightarrow 4 = t+2 \Rightarrow t = 2$ Hence, $\text{Area}(R) = \int_0^2 \frac{1}{(t+1)(t+2)} dt$	Must state $\frac{dx}{dt} = \frac{1}{t+2}$ $\text{Area} = \int \frac{1}{t+1} dx$. Ignore limits. $\int \left(\frac{1}{t+1} \right) \times \left(\frac{1}{t+2} \right) dt$. Ignore limits. changes limits $x \rightarrow t$ so that $\ln 2 \rightarrow 0$ and $\ln 4 \rightarrow 2$
(b)	$\left(\frac{1}{(t+1)(t+2)} \right) = \frac{A}{(t+1)} + \frac{B}{(t+2)}$ $1 = A(t+2) + B(t+1)$ Let $t = -1, 1 = A(1) \Rightarrow \underline{A = 1}$ Let $t = -2, 1 = B(-1) \Rightarrow \underline{B = -1}$ $\int_0^2 \frac{1}{(t+1)(t+2)} dt = \int_0^2 \frac{1}{(t+1)} - \frac{1}{(t+2)} dt$ $= [\ln(t+1) - \ln(t+2)]_0^2$ $= (\ln 3 - \ln 4) - (\ln 1 - \ln 2)$ $= \ln 3 - \ln 4 + \ln 2 = \ln 3 - \ln 2 = \ln\left(\frac{3}{2}\right)$	$\frac{A}{(t+1)} + \frac{B}{(t+2)}$ with A and B found Finds both A and B correctly. Can be implied. (See note below) Either $\pm a \ln(t+1)$ or $\pm b \ln(t+2)$ Both $\ln$ terms correctly ft. Substitutes <i>both</i> limits of 2 and 0 and subtracts the correct way round. $\frac{\ln 3 - \ln 4 + \ln 2}{\text{or } \ln\left(\frac{3}{4}\right) - \ln\left(\frac{1}{2}\right)}$ $\text{or } \frac{\ln 3 - \ln 2}{\text{or } \ln\left(\frac{3}{2}\right)}$ (must deal with $\ln 1$)

Question Number	Scheme	Marks
(c)	$x = \ln(t+2), \quad y = \frac{1}{t+1}$ $e^x = t+2 \Rightarrow t = e^x - 2$ $y = \frac{1}{e^x - 2 + 1} \Rightarrow y = \frac{1}{e^x - 1}$	Attempt to make $t = \dots$ the subject giving $t = e^x - 2$ M1 Eliminates t by substituting in y giving $y = \frac{1}{e^x - 1}$ A1 dM1 [4]
<i>Aliter</i> (c) Way 2	$t+1 = \frac{1}{y} \Rightarrow t = \frac{1}{y} - 1 \text{ or } t = \frac{1-y}{y}$ $y(t+1) = 1 \Rightarrow yt + y = 1 \Rightarrow yt = 1 - y \Rightarrow t = \frac{1-y}{y}$ $x = \ln\left(\frac{1}{y} - 1 + 2\right) \text{ or } x = \ln\left(\frac{1-y}{y} + 2\right)$ $x = \ln\left(\frac{1}{y} + 1\right)$ $e^x = \frac{1}{y} + 1$ $e^x - 1 = \frac{1}{y}$ $y = \frac{1}{e^x - 1}$	Attempt to make $t = \dots$ the subject M1 Giving either $t = \frac{1}{y} - 1$ or $t = \frac{1-y}{y}$ A1 Eliminates t by substituting in x dM1 giving $y = \frac{1}{e^x - 1}$ A1 [4]
(d)	Domain: $x > 0$	$x > 0$ or just > 0 B1 [1]
		15 marks

Question Number	Scheme	Marks
Aliter (c) Way 3	$e^x = t + 2 \Rightarrow t + 1 = e^x - 1$	Attempt to make $t + 1 = \dots$ the subject giving $t + 1 = e^x - 1$ M1 A1
	$y = \frac{1}{t+1} \Rightarrow y = \frac{1}{e^x - 1}$	Eliminates t by substituting in y giving $y = \frac{1}{e^x - 1}$ dM1 A1 [4]
Aliter (c) Way 4	$t + 1 = \frac{1}{y} \Rightarrow t + 2 = \frac{1}{y} + 1 \text{ or } t + 2 = \frac{1+y}{y}$	Attempt to make $t + 2 = \dots$ the subject Either $t + 2 = \frac{1}{y} + 1$ or $t + 2 = \frac{1+y}{y}$ M1 A1
	$x = \ln\left(\frac{1}{y} + 1\right) \text{ or } x = \ln\left(\frac{1+y}{y}\right)$ $x = \ln\left(\frac{1}{y} + 1\right)$ $e^x = \frac{1}{y} + 1 \Rightarrow e^x - 1 = \frac{1}{y}$ $y = \frac{1}{e^x - 1}$	Eliminates t by substituting in x giving $y = \frac{1}{e^x - 1}$ dM1 A1 [4]



Question 10

Question Number	Scheme	Marks
(a)	$\frac{dV}{dt} = 1600 - c\sqrt{h} \quad \text{or} \quad \frac{dV}{dt} = 1600 - k\sqrt{h},$ Either of these statements $(V = 4000h \Rightarrow) \frac{dV}{dh} = 4000$ $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt} = \frac{\frac{dV}{dt}}{\frac{dV}{dh}}$ $\text{Either, } \frac{dh}{dt} = \frac{1600 - c\sqrt{h}}{4000} = \frac{1600}{4000} - \frac{c\sqrt{h}}{4000} = 0.4 - k\sqrt{h}$ or $\frac{dh}{dt} = \frac{1600 - k\sqrt{h}}{4000} = \frac{1600}{4000} - \frac{k\sqrt{h}}{4000} = 0.4 - k\sqrt{h}$ Convincing proof of $\frac{dh}{dt}$	M1 M1 A1 AG
(b)	When $h = 25$ water leaks out such that $\frac{dV}{dt} = 400$ $400 = c\sqrt{h} \Rightarrow 400 = c\sqrt{25} \Rightarrow 400 = c(5) \Rightarrow c = 80$ From above; $k = \frac{c}{4000} = \frac{80}{4000} = 0.02$ as required	Proof that $k = 0.02$ B1 AG
<i>Aliter</i> (b) Way 2	$400 = 4000k\sqrt{h}$ $\Rightarrow 400 = 4000k\sqrt{25}$ $\Rightarrow 400 = k(20000) \Rightarrow k = \frac{400}{20000} = 0.02$	Using 400, 4000 and $h = 25$ or $\sqrt{h} = 5$. Proof that $k = 0.02$ B1 AG
(c)	$\frac{dh}{dt} = 0.4 - k\sqrt{h} \Rightarrow \int \frac{dh}{0.4 - k\sqrt{h}} = \int dt$ $\therefore \text{time required} = \int_0^{100} \frac{1}{0.4 - 0.02\sqrt{h}} dh \quad \div 0.02$ $\text{time required} = \int_0^{100} \frac{50}{20 - \sqrt{h}} dh$	<i>Separates the variables with</i> $\int \frac{dh}{0.4 - k\sqrt{h}}$ and $\int dt$ on either side with integral signs not necessary. M1 oe Correct proof A1 AG

Question Number	Scheme	Marks
(d)	$\int_0^{100} \frac{50}{20-\sqrt{h}} dh \quad \text{with substitution } h = (20-x)^2$ $\frac{dh}{dx} = 2(20-x)(-1) \quad \text{or} \quad \frac{dh}{dx} = -2(20-x) \quad \text{Correct } \frac{dh}{dx}$ $h = (20-x)^2 \Rightarrow \sqrt{h} = 20-x \Rightarrow x = 20-\sqrt{h}$ $\int \frac{50}{20-\sqrt{h}} dh = \int \frac{50}{x} \cdot -2(20-x) dx$ $= 100 \int \frac{x-20}{x} dx$ $= 100 \int \left(1 - \frac{20}{x}\right) dx$ $= 100(x - 20\ln x) (+c)$ change limits: when $h=0$ then $x=20$ and when $h=100$ then $x=10$ $\int_0^{100} \frac{50}{20-\sqrt{h}} dh = [100x - 2000\ln x]_{20}^{10}$ or $\int_0^{100} \frac{50}{20-\sqrt{h}} dh = [100(20-\sqrt{h}) - 2000\ln(20-\sqrt{h})]_0^{100}$ $= (1000 - 2000\ln 10) - (2000 - 2000\ln 20)$ $= 2000\ln 20 - 2000\ln 10 - 1000$ $= 2000\ln 2 - 1000$ Correct use of limits, ie. putting them in the correct way round Either $x=10$ and $x=20$ or $h=100$ and $h=0$ Combining logs to give... $\begin{aligned} &2000\ln 2 - 1000 \\ &\text{or } -2000\ln\left(\frac{1}{2}\right) - 1000 \end{aligned}$	B1 aef M1 M1 A1 ddM1 A1 aef [6]
(e)	Time required = $2000\ln 2 - 1000 = 386.2943611... \text{ sec}$ = 386 seconds (nearest second) = 6 minutes and 26 seconds (nearest second) <u>6 minutes, 26 seconds</u>	B1 [1]
		13 marks